ÖZYEĞİN UNIVERSITY

Course Syllabus
EE 321 Microprocessors
SPRING 2025-26

INSTRUCTOR	
Name	*H. Fatih Uğurdağ*
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Phone Number	*0216 564 94 36*
Office Number	*AB1-212*
Office Hours	*Wednesday 14:30-15:30*
Teaching Assistants (TAs)	*Head TA: C. Emre Dedeağaç (AB1-226)* *Grad TAs: Seçkin İpek (226), T. Azhar Shah (226)* *Undergrad TAs: Arda Güler, Mert Kırgın*

COURSE ORGANIZATION	
Interaction Methods	*Lectures, Piazza, Office hours*
Technology Used by the Instructor	*LMS, Piazza, Web, Vivado, Zoom*
Technology Requirements for Students	*Computer with Vivado and Zoom installed*

COURSE MATERIALS	
Required Books	-
Recommended Book	*Programming 8-bit PIC Microcontrollers in C, Martin P. Bates, Newnes.*
Other Materials	*ugurdag.com/ee321, LMS, Piazza*
Accessing Course Materials	*Through a computer*

WEEKLY SUBJECTS (Subject to Change)		
Week	**Date**	**Subject**
1	Feb 11	*Intro*
2	Feb 18	*VerySimpleCPU (VSCPU) Architecture*
3	Feb 25	*VSCPU Assembly Programming – General Concepts (HW1)*
4	Mar 4	*VSCPU Assembly Programming – Loops and Arrays (HW2)*
5	Mar 11	*VSCPU Assembly Programming – Function Calls (HW3, **Lab1**)*
6	Mar 18	*Arch. Concepts + PIC MCU + PIC Assembly (**Lab2**)*
7	Mar 25	*PIC + Banking + ADC + GPIO + Various Peripherals (**Lab3**)*
8	Apr 1	*Banking **+ Project** Kick-off (**Lab4**)*
9	Apr 8	***Midterm***
10	**Apr 15 (Withdrawal week)**	*Datasheets + Timers and Interrupts + MCU Overview (Quiz1, **Lab5**)*
11	Apr 22	*PWM + Serial Communications*
12	Apr 29	*Advanced Computer System Subjects and uPs (Quiz2)*
13	May 6	*Advanced Computer Sys. Subjects – Cont'd (**Lab6**)*
14	May 13	*Advanced Computer Sys. Subjects – Cont'd (Quiz3)*
15	May 20	***Project** Demo in the Lab*

ASSESSMENT METHODS, WEIGHTS AND RULES			
Type	**Weight**	**Implementation Rule**	**Makeup Rule**
Final Exam	*20%*	*Closed book and notes. No calculation or communication devices will be allowed during the exam.*	*BÜT exam will serve as the makeup for the final exam.*
Midterm Exam	*16%*	*Closed book and notes. No calculation or communication devices will be allowed during the exam.*	*Only valid excuses with an official report are accepted to qualify for a midterm makeup..*

Homeworks	*9%*	*Copying the work of others or AI is not permitted. There will be 3 HWs.*	*There will be no makeup for missing a HW. Late HWs is accepted with penalty.*
Lecture Attendance	*6%*	*Less than 30 hours fails the course.*	*Only valid excuses with an official report are accepted.*
Quizzes	*9%*	*3 Quizzes.*	*No make-up.*
Project	*20%*	*TBD.*	*No late submissions are accepted.*
Labs	*20%*	*If you miss more than 2 labs, you get zero for your total lab score.*	*The lab with the lowest score is excluded. That is your also make-up.*
Total	**100%**		

METHODS FOR TRACKING LECTURE ATTENDANCE & ITS ON SUCCESS			
Method	**Application Details**	**Required Tools and Make-up Method**	**Impact of Lecture Attendance on Course Grade**
QR Code (MyOzU)	*At the beginning and the end of class, a QR code will be displayed on the screen.* *Each scanning of the QR (beg. or end) with MyOzU app on your cell phone will be counted as 1.5 hours.* *You can access the guideline here*	*You should have a cell phone with MyOzU app installed.* *If you miss class and provide an official report with a valid excuse, you will be treated as if you have attended the class.* *You will also be able to collect more attendance points when attending problem sessions (through signature sheets).* *On Feb 9, everybody gets full credit.*	*Less than 30 hours fails the course.* *You will collect 0.25 point for every hour of attendance up to a max of 10 points.*

ASSESSMENTS BEFORE THE WITHDRAWAL DEADLINE		
Type of Assessments	**Weights**	**Date by When the Score of the Assessment is to be Announced**
Midterm	*16%*	*April 11-15*
Homeworks	*9%*	*April 11-15*
Labs (partial)	*13.3%*	*April 11-15*
Lecture Attendance (partial)	*3.6%*	*April 11-15*

DETERMINING THE LETTER GRADE
Catalog *90-100: A, 85-89: A-, 80-84: B+, 75-79: B, 70-74: B-, 66-69: C+, 62-65. C* *58-61: C-, 54-57: D+, 50-53: D, 0-49: F*

EXPECTED STUDENT SEMESTER WORKLOAD	
Activities Carried Out in the Presence of An Instructor	**Total Hours in Semester**
Lectures	*45*
Labs	*18*
Activities Carried Out by the Learner Him/Herself	**Total Hours in Semester**
Homeworks + Lab prep	*27*
Review of Course Material for Test Preparation	*25*
Project	*40*
Overall Total Hours in Semester	***155***

RULES FOR THE USE OF AI IN THE COURSE

Use of AI in the Course **(Free Use within Ethical Framework)**	You may use AI tools (ChatGPT, Copilot, Gemini, Claude, Grok, DeepSeek, etc.) for the project only, provided that you do so **ethically** and in compliance with the **Principles for the Responsible Use of AI** listed below. You may benefit from AI in assignments and other learning activities. However, you must clearly acknowledge how you used AI, act ethically and transparently, and comply with university policies. *The key is that you own what you submit. It is your solution whether AI's contribution is 100%, 50%, or 0%. You should be in the driver's seat and be able to test, debug, fix, present, and defend your submitted solution. The goal is that you come up with a unique/original design and test suite.*
Principles for the Responsible Use of AI	1. **Use AI as a tool to support your learning, not as the ultimate source of knowledge:** AI can assist your learning and work, but the final judgment, creativity, and ownership of the output must always remain with you. 2. **Be transparent about AI's contribution:** If you use AI to generate, summarize, or improve any part of your work, you must clearly acknowledge it. Transparency protects your academic honesty and strengthens professional integrity. 3. **Recognize the limitations and biases of AI:** AI may produce inaccurate, incomplete, or biased outputs. Always review the results critically and verify them before using. 4. **Consult diverse sources, not just AI:** Do not rely solely on a single AI. Support, compare, and cross-check AI outputs with other AIs and other valid sources. 5. **Protect your data and privacy:** Never share personal, sensitive, or confidential information with AI tools. Safeguarding such information is your responsibility.

OTHER RULES AND INFORMATION	
Health	The mode of instruction is subject to change in case of pandemic or other similar unavoidable reasons in line with the directives of the authorized public bodies. In such a case, the necessary revisions will be binding for all members of the OzU community and will be announced by the relevant units, in particular Student Services.
Safety	Announced safety procedures must be strictly followed during any laboratory and similar work. Please refer to the safety guidelines posted in the laboratories. In case of emergencies, call 9911 (216 564 9911).
Accessibility	Reasonable accommodations will be provided for students with verifiable disabilities. Please inform your instructor if you need any assistance.
Academic and Technical Support	Students can get help from the Writing Center, Solution Center and IT in case they need academic and/or technical support with their classes.
Objections to Examination Results	Students who object to their exam results and cannot resolve their objections with the instructor of the course, may file a written petition for a review of their exam papers at their Dean's/Director's Office within five business days from the announcement of their exam results.
Academic Integrity and Honor Code	Students are assumed to abide by scholastic honesty and uphold the Honor Code. Scholastic dishonesty includes, but is not limited to cheating, attempting to cheat, plagiarizing, fabricating information or citations, facilitating acts of dishonesty by others, having unauthorized possession of examinations, submitting the work of another person, or tampering with the academic work of other students. Any form of scholastic dishonesty is a serious academic violation and will result in a disciplinary action and also may result in student getting a 0 grade in the related assessment.
Unauthorized Recording/Sharing of Lectures	Recording/sharing of synchronous/asynchronous lectures or copying of lecture recordings are prohibited without the permission of the instructor. Actions such as unauthorized recording, giving the recording to someone else, taking it, using it or sharing it in physical or virtual media are prohibited and/or are offenses according to the relevant legislation. In such cases, legal procedures will be initiated in addition to starting disciplinary actions against those involved.
Flexibility	Circumstances may arise during the course that prevent the instructor from fulfilling each and every component of this syllabus; therefore, the syllabus is subject to change. Students will be notified prior to any changes.
Principles for Classroom Behavior	Creating a safe space where ideas can be freely discussed in our classrooms is everyone's responsibility. In accordance with the relevant disciplinary regulations, any verbal or written expression that disrupts the order in classroom, that involves unauthorized distribution of materials, that is discriminatory/violent, or that violates dignity of individuals in the classroom (students or educators) is strictly prohibited. During class, all participants are expected to respect each other's rights.